GEORGIA INSTITUTE OF TECHNOLOGY

ECE 6256 / CS 6604 Conversational AI Fall 2026

Assignment #0 — Survey Article

Assigned: Sep 1st 2026
Due Date: 11:59pm Sep 7th 2026

Please use Piazza to clarify the instructions for the project assignments.

This assignment aims to create an AI Virtual Assistant (AVA) to assist scientists in writing scientific survey articles.[1] For this assignment:

- Use the web interface for Meta AI (https://www.meta.ai/)
- This uses Meta's Muse Spark 1.2. For instructions (e.g., how to create your own prompt), see https://www.meta.com/help/artificial-intelligence/change-meta-ai-model/?ref=learn_more

  This model is free.

Part 1. AVA Chatbot for Writing Scientific Surveys

A chatbot that interacts with a user to assist them in writing and formatting the survey article.

- Design prompts so that the chatbot acts as an AVA to assist you in writing a scientific survey article[2]. You may need to iterate with AVA and perhaps even manually fix certain aspects of the paper, add more to a section, less to others, etc. Ultimately, this is about using AVA to help you write *your* best paper - not an automated paper writer. **Your paper will be graded on its quality.**
- The survey paper should be an ACL Conference Long Paper[3] (8 pages). Your last prompt should tell the chatbot to initiate a session by asking the user
  "Which topic would you like to write a survey?"
- After iterative refining your prompts to improve the quality of the survey paper, save (copy and paste) the final prompts to the file :
  "AVAScholarPrompts_{your GT user ID}.txt".
- Once you have completed your prompt designs, start a new session, copy and paste the prompts from "AVAScholarPrompts_{your GT user ID}.txt", and use your AVA to assist you in writing a scientific survey article on the topic of "Conversational AI".
- (Manually) input the text of the survey article into the LaTeX template for an ACL Conference Long Paper. You should use this template on Overleaf:
  https://www.overleaf.com/latex/templates/association-for-computational-linguistics-acl-conference/jvxskxpnznfj

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[1] If you choose to submit a paper written with the help of an AVA to a conference, please follow the AI Writing Assistants Policy here: https://aclrollingreview.org/cfp

[2] Later in the course, we will download LLMs and use more advanced methods including fine-tuning, Retrieval Augmented Generation (RAG), etc. but for now, develop your AVA with prompts only.

[3] Association for Computational Linguistics.

- Generate and download a PDF of the paper from Overleaf. The PDF should adhere to the paper formatting guidelines of the ACL, as implemented in the Overleaf template above (e.g., 8 pages not counting the References). Save the PDF as **"ConversationalAISurvey_{your GT user ID}.pdf"**.

- Capture the text of the conversational interaction with the prompt "Copy all the conversational interactions of this session with the user messages denoted by "User:" and the AI responses denoted by "AI:".

- Save (copy and paste) the text of the conversational interaction to a file **"AVAConvInteractions_{your GT user ID}.txt"**.

Part 2. AVA Scientific Article Assessment
Critical assessment of scientific articles

- Design prompts to create an AVA that reviews a scientific article. The review should follow the guidelines of the ACL Review[4] *except that the final score should be on a scale from 0 to 100 points, with 100 as the best score*.

- The final prompt should ask the user to enter the text of the paper.

- Save (copy and paste) the prompts into a file **"AVASciPaperReviewerPrompts_{your GT user ID}.txt"**.

- Use your AVA to assess the survey paper on Conversational AI completed in Part 1 above. To accomplish this:

  1. Start a new session in Meta AI
  2. Copy and paste the prompts from "AVASciPaperReviewerPrompts_{your GT user ID}.txt" into Meta AI
  3. Copy and paste the text of your survey article into Meta AI
  4. Use the criticisms to improve your paper in Part 1, rerun Part 1, and generate a new (improved) survey paper.
  5. Re-run Steps 1-3.
  6. Check that the review and the final score have improved.
  7. Save the review into the file **"AVASciPaperReview_{your GT user ID}.txt"**.

Part 3. AVA Chatbot Assessment
Evaluate the quality of your AVA Chatbot for Writing Scientific Surveys.

- Design prompts to assess the quality of the conversational AI system you designed in *Part 1. AVA Chatbot for Writing Scientific Surveys*. The assessment should use state-of-the-art evaluation metrics of chatbots [5]. The assessment should include a score. The score should be between 0 and 100, where 100 is the best possible score.

- The final prompt should ask the user to enter the conversational interaction.

[4] https://aclrollingreview.org/reviewertutorial

[5] Refer to your survey article on Conversational AI that your AVA helped you write to determine the Conversational metrics used to measure a conversational AI system.

- Save the prompts into a file **"AVAChatbotAssessmentPrompts_{your GT user ID}.txt"**.
- Use your AVA to assess the quality of your AVA Scientific Survey Article Chatbot from Part 1 above. To accomplish this:
    1. Start a new session in Meta AI
    2. Copy and paste the prompts from "AVAChatbotAssessmentPrompts_{your GT user ID}.txt" into Meta AI
    3. Copy and paste the conversational interaction (see Part 1) from "AVAConvInteractions_{your GT user ID}.txt" into Meta AI
    4. Save the assessment into the file **"AVAChatbotAssessment_{your GT user ID}.txt"**.

To complete the assignment, zip these files into one file and upload it on Canvas Assignment 0:

1. **"AVAScholarPrompts_{your GT user ID}.txt"**
2. **"ConversationalAISurvey_{your GT user ID}.pdf"**
3. **"AVAConvInteractions_{your GT user ID}.txt"**
4. **"AVASciPaperReviewerPrompts_{your GT user ID}.txt"**
5. **"AVASciPaperReview_{your GT user ID}.txt"**
6. **"AVAChatbotAssessmentPrompts_{your GT user ID}.txt"**
7. **"AVAChatbotAssessment_{your GT user ID}.txt"**